

Design shipTRACKer database for SOTA company with the following data requirements:

1. SOTA company would like to manage all ships related to SOTA's transactions. Each ship is uniquely named and has recorded design drawings. A ship can be owned by a person or other company. A ship owner is recorded with a unique code, contact detail including phone number and address, and business domain; however, for each owner type, SOTA requires more different details, e.g. for each person, SOTA needs his/her social security number and job title and for each company, SOTA needs its unique trade mark and commercial registration.
2. The type of each ship is also important to SOTA so that SOTA can define various services suitable for each ship type. In SOTA, many ships are known of the same type; however, there are also some new ship types recorded for future ships in SOTA's business. A ship type has a unique name, tonnage, and hull. The key designer of each ship type is also of SOTA's interest for further communication about ship's maintenance services.
3. Each ship has many workers. Each worker is uniquely identified by his/her social security number, has first name, last name, and experiences. For different tasks on each ship, workers work in three different groups: control, operation, and supporting. The captain and his/her sailing leaders belong to control group, other sailors belong to operation group, and the remaining workers belong to supporting group.
4. It is worth mentioning that the captain and his/her sailing leaders are experienced sailors. A ship must be managed by one captain and the captain manages a ship in a particular period of time. Each other sailor is also recorded with his/her working periods of time on his/her current ship.
5. Each ship has its own log showing its movements. Each movement is uniquely recorded via logged timestamp including date and time, in addition to its location including longitude and latitude. In ship voyages, SOTA also wants to capture the details of the ports that each ship visited. Each port visit in many visits of a particular ship at a port is recorded uniquely with the starting date. The end date of each visit is also known. For each port visit, the ship may use SOTA's services at different agreement levels. A service may be used by many different ships or even the same ships in their visits. Each service is uniquely named and has a short description and use contexts.
6. Each port is uniquely named in its country, which is uniquely identified by either code or name in a particular main continent. A port might be a home port of many ships; however, some ports may not. A ship is registered with only one home port if needed. Its registered date for a home port is also captured.
7. Each port is located on a sea, ocean, or lake, each of which is uniquely identified by a global name. Their weather conditions are also published for all visitors. A sea, ocean, or lake can have many ports.